

Edge Addition Planarity Suite and Generalized Graph Library

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Summary

A *graph* is a mathematical structure comprising a set V of vertices and a set E of edges, with each edge e corresponding to a pair of vertices called the *endpoints* of e . A graph is *planar* if its vertices can be placed in distinct locations on a plane surface and its edges can be drawn on the plane without the edges intersecting, except at their common vertex endpoints. For a planar graph, a planarity algorithm typically outputs a combinatorial data structure that validation code can use to certify the planarity of the input graph. A planar graph drawing is typically produced by a separate algorithm. On the other hand, if an input graph is not planar, then a planarity algorithm typically outputs a minimal subgraph of the input graph that obstructs planarity. Validation code can use a minimal planarity-obstructing subgraph to certify the non-planarity of an input graph. Moreover, a minimal subgraph obstructing planarity can be used to help decide how to amend an input graph to *planarize* it.

Graphs are used to model a very wide array of real-world problems in which there are objects and relationships between the objects. In artificial intelligence, graphs are used to help with reasoning tasks, such as about the relationship pathways between persons of interest in law enforcement or between genes, tissues, diseases, and medications in bioinformatics research. In physics, graphs and planarity are used to help compute material phase transitions, particle interactions, and electromagnetic duality. Similarly, in chemistry, graphs may be used to represent atoms and their valence bonds in molecules. Planarity testing can help determine feasible molecular arrangements because the molecular graphs for many compounds of interest must be planar. In electronics, a graph may be used to represent the electrical components and wiring in a circuit, such as transistors and etchings on a silicon wafer. The graph of a circuit must be planar or planarized to eliminate short circuits.

The open source software described in this paper provides a highly performant generalized graph library that also implements state-of-the-art algorithms for planarity-related problems. The generalized graph library and planarity-related algorithms are available to C/C++ and Python graph application developers.

Statement of need

The Edge Addition Planarity Suite is an open source software package that was originally developed to implement a new planarity algorithm in Boyer & Myrvold (2004). The Edge Addition Planarity Algorithm is currently regarded as one of two state-of-the-art planarity algorithms due to its conceptual simplicity as well as its speed of execution (Contributors to Wikipedia, 2026). The software package is available from the [edge-addition-planarity-suite GitHub repository](#) under a 3-clause BSD license.

To enable development of the Edge Addition Planarity Algorithm, as well as the other planarity-related algorithms in Boyer (2006) and Boyer (2012), it was necessary to develop an extensible,

41 generalized graph library. This generalized graph library provides a high-performance solution
42 for the graph algorithm representation and processing needs of a very wide array of graph
43 applications.

44 In addition to enabling graph application development in C/C++, the popularity of Python as
45 a scientific application development language has prompted the need for making this
46 graph library and the algorithms it implements available to Python developers. The source code
47 may be accessed by the [planarity Github repository](#), and the [planarity Python package](#) may
48 be installed via the Python Package Index (PyPI) by simply running `pip install planarity`.

49 State of the field

50 Several large scientific software packages offer a graph planarity algorithm, which is a complex
51 graph algorithm suitable for benchmarking graph libraries. Mathematica and Wolfram Alpha
52 include planarity algorithms, but these packages are not meant for high-performance graph
53 algorithms, so the Library for Efficient Data Types and Algorithms ([LEDA](#)) is significantly
54 faster than they are because it is an industrial library designed for computational efficiency. Yet,
55 in Boyer et al. (2004), all planarity algorithms implemented in LEDA were shown to be several
56 times slower than the Edge Addition Planarity Algorithm implementation. LEDA also has the
57 disadvantage of being C++ only and has a proprietary license. The only graph library shown
58 (in Boyer et al. (2004)) to have similar performance to the Edge Addition Planarity Suite is the
59 Public Implementation of a Graph Algorithm Library and Editor ([PIGALE](#)). However, PIGALE is
60 also C++ only, has a GPLv2 license, and lacks advanced algorithms for homeomorphic subgraph
61 search. For example, torus embedding algorithms and torus obstruction isolation algorithms
62 are able to operate differently based on whether a search for a subgraph homeomorphic to
63 $K_{3,3}$ finds a result (Myrvold & Kocay, 2011; Myrvold & Woodcock, 2018). Similarly, several
64 graph theoretic results are applicable to graphs that are known to be $K_{3,3}$ -free, including
65 results related to graph isomorphism, reachability, and finding matching cuts (Datta et al.,
66 2009; Feghali et al., 2025; Thierauf & Wagner, 2014).

67 Software design

68 The generalized graph library in the Edge Addition Planarity Suite has a carefully curated API
69 and an object-oriented architecture. The base **Graph** class structure includes

- 70 ■ base graph, vertex, and edge structure definitions and getter/setter methods;
- 71 ■ graph, vertex and edge allocation, deallocation, and reset methods;
- 72 ■ methods for iterating through all vertices, edges, and edge adjacency lists;
- 73 ■ graph readers and writers for several formats; and
- 74 ■ a dynamic subclassing/extension mechanisms and virtual function table.

75 The base Graph is extendable with utility methods for exploring a graph, labelling vertices and
76 edges according to a depth-first search (DFS), and for managing connectivity and biconnectivity
77 information. The vertex and edge structures are extended to enable storage of the information
78 generated by the utility methods.

79 A Graph structure extended with the DFS-related utilities is further extendable to a **Planarity**
80 **Graph**, which also further extends the vertex and edge structures. The main additional method
81 provided by a Planarity Graph is `gp_Embed()`, which takes a Planarity Graph as input and
82 outputs either a combinatorial planar embedding or a minimal planarity-obstructing subgraph.

83 Given a Planarity Graph, a number of subclass extensions are available for planarity-related
84 problems. One extension solves outerplanarity. Another implements the planar graph drawing
85 algorithm in Boyer (2006). Three other extensions provide Graph subclasses that solve the
86 homeomorphic subgraph search algorithms appearing in Boyer (2012). All of these extensions
87 further extend the vertex and edge structures and overload the `gp_Embed()` function.

88 In addition to being available to C and C++ developers, the APIs of the Edge Addition
89 Planarity Suite and Generalized Graph Library are also made available via the planarity
90 Python package with the source code made available in the [planarity Github repository](#).
91 Cython is used to preserve high performance while also broadening availability to the Python
92 developer community.

93 A final aspect of this open source software project is our emphasis on software reliability. We
94 perform validation code on `gp_Embed()` and its five overloads on billions of randomly generated
95 graphs up to 10 million vertices and 30 million edges. The GitHub actions that run on every
96 pull request include sanitizer tests on all graphs on 8 vertices. Finally, the [edge-addition-
97 planarity-suite-testing Github repository](#) provides our multithreaded Python code for
98 validating `gp_Embed()` and its five overloads on the more than one billion graphs having 11 or
99 fewer vertices, as generated by the geng tool in Nauty and Traces ([McKay & Piperno, 2025](#)).

100 Research impact statement

101 The journal publication of the Edge Addition Planarity Algorithm ([Boyer & Myrvold, 2004](#))
102 currently has over 360 scholarly citations according to [Google Scholar](#). The algorithm has
103 been implemented in several large open source projects, including [Boost](#), [Magma](#), [nauty](#) and
104 [Traces](#) ([McKay & Piperno, 2025](#)), and the [Open Graph Drawing Framework](#) ([web archive](#)).
105 An executable program built from the source code was used to build a knot theory software
106 package ([Jablan & Sazdanović, 2007, pp. 7, 38](#)). The source code from the Edge Addition
107 Planarity Suite repository has been directly included in several large open source projects,
108 including [SageMath](#), [Digraphs](#), and over 80 Linux platforms releases including for Debian,
109 Devuan, Kali, openSUSE, Raspbian, Ubuntu, Alpine, ALT, Fedora, FreeBSD, Gentoo, Manjaro,
110 MSYS2, and Parabola, according to [repology/edge-addition-planarity-suite](#) ([web archive](#)) and
111 [repology/planarity](#) ([web archive](#)).

112 The Python package named planarity was originally developed in 2013 by Aric Hagberg as a
113 way to get the planarity testing and drawing algorithms from Boyer & Myrvold ([2004](#)) and
114 Boyer ([2006](#)) to work with graphs from NetworkX ([Hagberg et al., 2008](#)). Recently, Hagberg
115 transferred the [planarity Github repository](#) to the [Github Graph Algorithms Organization](#)
116 and the [planarity Python package](#) to the [PyPI Graph Algorithms Organization](#), which are
117 both maintained by the authors. On August 1, 2026, the [Top PyPI Packages](#) website indicated
118 that the planarity Python package had over 700,000 downloads for the preceding month and
119 was ranked in the top 1% of PyPI package downloads (ranked 5262 out of [864,036 projects](#)).
120 According to [pepy.tech](#) ([web archive](#)) the planarity Python package has been downloaded
121 over 2 million times in total. The planarity Python package has now been included as a
122 standard Python package in several versions of Linux including Debian, Devuan, Kali, Raspbian,
123 and Ubuntu, according to [repology/python:planarity](#) ([web archive](#)).

124 AI usage disclosure

125 No generative AI tools were used in the writing of this manuscript, nor in the preparation
126 of supporting materials. The authors are not aware of any substantive use of generative AI
127 tooling yet in the development of the open source software projects reported in this paper.
128 However, in light of the proliferation of availability and use of generative AI tooling in software
129 development, we assert that it is our pull request (PR) review process, performed with no
130 generative AI tooling, that protects the projects and their consumers from low quality or
131 harmful contributions. As the project owners, we amend PRs with further commits or request
132 contributor changes as needed, and we use our own holistic project expertise to ensure that
133 the code contribution meets our quality requirements. Specifically, a code contribution must
134 cover all parts of the project that it should, its code must be consistent with the rest of the
135 code in the module(s) containing it, it must be a good solution conforming to every aspect

of the technical specifications in the issue(s) it solves, the tests also given in the technical specifications must be correctly coded, the coded solution must pass the coded tests, and all code in the contribution must pass our sanitizer and compiler-warning-sensitive automated tests. Only once a project owner has determined that these requirements are met by the pull request's code contribution is the pull request approved to be merged.

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